

EAS 3530 – Space Systems Concepts

Project 1

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Section 1: Mission Concept

NASA's TechPort website lists project ID 112843, which essentially acts as a lunar rover. We also see that NASA recognizes a technology gap with the ability to "provide adequate food, water, and supplies to ensure crew mental and physical wellbeing during long duration missions." Within this topic, we believe that a gap could be closed by using a rover landed on the moon to test and categorize the chemical makeup of the lunar soil to see what types of plants we could grow on the lunar surface, with the ability to send the collected regolith back to the Earth. With this, a basic mission concept can be put together.

The mission will aim to land an autonomous rover on the lunar surface, have that rover collect samples of the lunar soil from different areas of interest, and be able to mix that soil with easy-to-obtain stimulants such as organic waste, and plant a seed in that mixture under a controlled environment to see if growing plants in prepared lunar soil is a viable option. With this in mind, the minimum success criteria are:

1. Successfully land the rover on the lunar surface without damage to the planting system or stimulants;
2. Collect a lunar sample at the landing site; and
3. Mix the collected lunar soil with the stimulants and place a seed in the soil.

The full success criteria include the minimum success criteria AND:

1. Demonstrate that plants can grow in lunar soil to maturity when stimulated by organic matter;
2. Collect lunar soil samples from 5 different locations in a 50-kilometer radius from the landing site; and
3. Place one of five different varieties of seeds in each of the samples collected.

Section 2: Requirements Definition and CONOPS

Mission Requirements

- **MR – 1: Autonomous System Traversal**
 - The mission shall deploy a rover capable of autonomous lunar terrain traversal
- **MR – 2: Payload Transport and Deployment**
 - The rover shall transport science or ISRU payloads of up to 5 kg across the lunar surface
- **MR – 3: Terrain Mapping and Data Collection**
 - The mission shall characterize lunar regolith properties relevant to plant growth
- **MR – 4: Communication and Control**
 - The rover shall achieve traverse speeds ≥ 1 m/s in lunar terrain

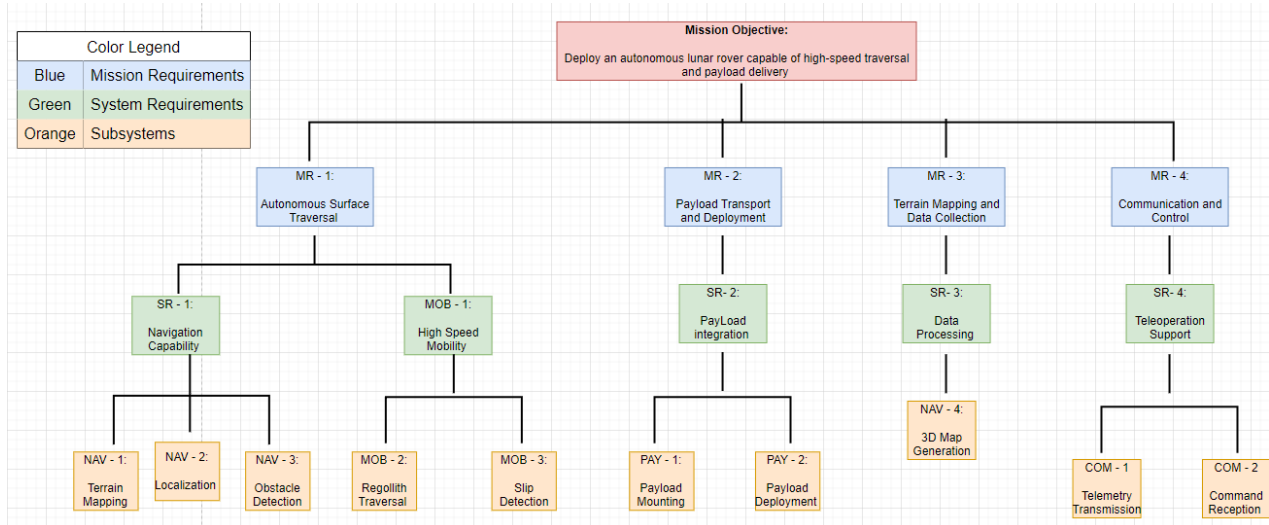
System Requirements

- **SR – 1: Navigation Capability**
 - The rover shall autonomously navigate unprepared terrain for up to 24 hours
- **SR – 2: Payload Integration**
 - The rover shall autonomously avoid terrain hazards
- **SR – 3: Data Processing**
 - The rover shall support human teleoperation
- **SR – 4: Teleoperation support**
 - The rover shall store and transmit up to 2 GB of terrain mapping data at a rate of 500 Kbps

Subsystem Requirements

- **NAV – 1: Terrain Mapping**
 - Navigation system shall generate 3D terrain maps
- **NAV – 2: Localization**
 - Navigation system shall localize rover position within 1 meter precision
- **NAV – 3: Obstacle Detection**
 - Navigation system shall detect obstacles larger than 5 cm.
- **NAV – 4: 3D Map Generation**
 - Navigation system shall support autonomous navigation for 24 hours
- **MOB – 1: High Speed Mobility**

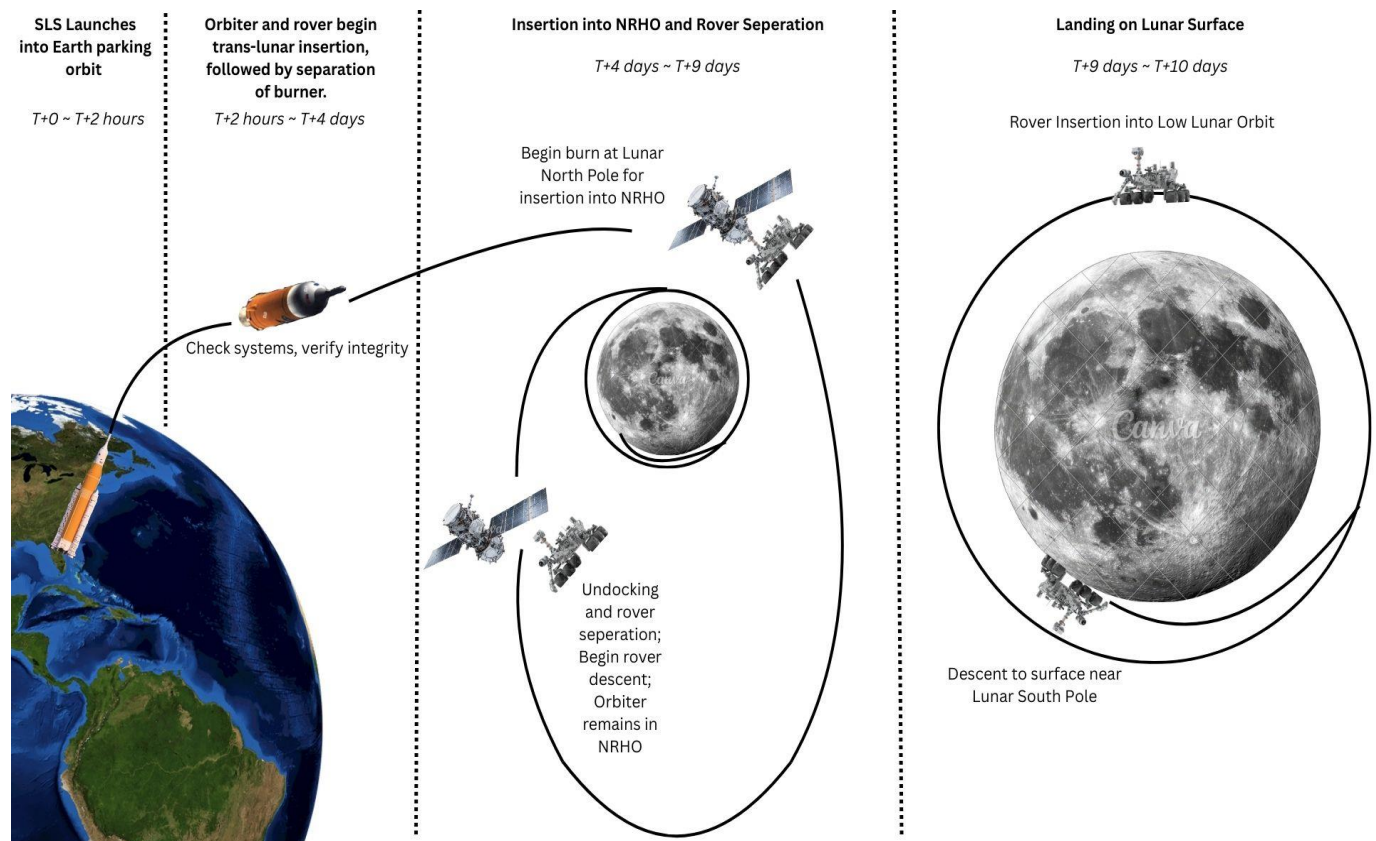
- Mobility system shall support rover traversal at ≥ 1 m/s
- **MOB – 2: Regolith Traversal**
 - Mobility system shall be able to operate on lunar terrain for 60 days without breakdown
- **MOB – 3: Slip Detection**
 - Mobility system shall detect wheel slippage
- **MOB – 4:**
 - Mobility systems shall be able to traverse obstacles smaller than 5 cm.
- **PAY – 1: Payload Mounting**
 - Payload system shall carry science instruments up to 8 kg
- **PAY – 2: Payload Deployment**
 - Payload system shall deploy instruments on the lunar surface
- **COM – 1: Telemetry Transmission**
 - Communication system shall transmit telemetry at a rate of 1 Mbps to ground control
- **COM – 2: Command Reception**
 - Communication system shall be able receive command signals at rates of 1 Mbps



Req ID Requirement			Verification Method
Req ID	Requirement	Level	Method

MR - 1	The mission shall deploy a rover capable of autonomous lunar terrain traversal	Mission	Demonstration
MR - 2	The rover shall transport science or ISRU payloads of up to 5 kg across the lunar surface	Mission	Test
MR - 3	The mission shall characterize lunar regolith properties relevant to plant growth	Mission	Demonstration
MR - 4	The rover shall achieve traverse speeds ≥ 1 m/s in lunar terrain	Mission	Field Test
SR - 1	The rover shall autonomously navigate unprepared terrain for up to 24 hours	System	Simulation
SR - 2	The rover shall autonomously avoid terrain hazards.	System	Test
SR - 3	The rover shall support human teleoperation	System	Demonstration
SR - 4	The rover shall store and transmit up to 2 GB of terrain mapping data at a rate of 500 Kbps	System	Test
MOB - 1	Mobility system shall support rover traversal at ≥ 1 m/s	Subsystem	Field Test
MOB - 2	Mobility system shall be able to operate on lunar terrain for 60 days without breakdown	Subsystem	Test
MOB - 3	Mobility system shall detect wheel slippage	Subsystem	Test
MOB - 4	Mobility system shall be able to traverse obstacles smaller than 5 cm.	Subsystem	Demonstration
NAV - 1	Navigation system shall generate 3D terrain maps	Subsystem	Simulation
NAV - 2	Navigation system shall localize rover position within 1 meter precision	Subsystem	Analysis

Navigation system shall detect obstacles larger NAV - 3 than 5 cm.	Subsystem	Test
Navigation system shall support autonomous NAV - 4 navigation for 24 hours	Subsystem	Simulation
Communication system shall transmit telemetry COM - 1 at a rate of 1 Mbps to ground control	Subsystem	Test
Communication system shall be able receive COM - 2 command signals at rates of 1 Mbps	Subsystem	Demonstration
Payload system shall carry science instruments PAY - 1 up to 8 kg	Subsystem	Inspection
Payload system shall deploy instruments on the PAY - 2 lunar surface	Subsystem	Demonstration



Mission Timeline:

Phase 1 — Launch

Launch vehicle places spacecraft in translunar injection.

Phase 2 — Lunar Orbit Insertion

Orbiter performs burn to enter NRHO.

Phase 3 — Lander Deployment

Lander separates from orbiter.

Lander enters low lunar orbit

Lander descends to lunar surface.

Phase 4 — Surface Operations

Rover deploys and begins experiments.

Phase 5 — Science Operations

- Regolith collection
- Soil treatment
- Seed planting
- Growth monitoring

Phase 6 — Data Transmission

Science data sent to orbiter which relays to Earth.

Section 3: Mission Geometry and Architecture

The experiment would require containers capable of storing both the seed and the organic material separately until the lunar soil sample is collected. It would then have to mix the organic matter with the soil, and lastly drop the seed into the soil, and do all of this while having a pressurized environment capable of withstanding the vacuum of space. A similar, though not as flexible, experiment has been performed on the lunar surface with China's Chang'e 4 lunar lander, which brought with it the Lunar Micro Ecosystem (LME), a sealed biosphere to the lunar surface. Although it was cut short due to a design oversight with the temperature regulation, is

showed what kinds of organisms and materials can be used to create a self-sustaining environment on the moon. The experiment here would require an enclosure similar to the LME, but with the capability of adding lunar soil to the environment.

Since NASA's LCRNS does not exist yet, the rover would have to communicate with the orbiter that landed it. In order to do this, the rover would have to undock from the orbiter and be able to land separately while the orbiter inserts itself into its orbit.

The lunar orbit will be an NRHO orbit with a 9:2 resonance with the lunar synodic period. This was chosen because of the minimal thrust that is required to keep the orbiter in its orbit, and because of the ability to maintain near-constant communications with ground stations on the Earth. It also has the added benefit of keeping high communication rates with the rover when it is situated near the southern lunar pole as it dwells above the area for a long time during its apoluna.

The orbit will have a perilune of about 1600 kilometers and an apolune of 69500 kilometers as reported by FreeFlyer. With this, we can find the semimajor axis $a = \frac{r_a + r_p}{2}$

and use that to find the period $T = 2\pi \sqrt{\frac{a^3}{\mu_{moon}}}$, where $\mu_{moon} = 4902.8 \frac{\text{km}^3}{\text{s}^2}$.

This gives a semimajor axis of 35,550 km and a period of 167.07 hours. The velocity at perigee can be found with $v_p = \sqrt{\mu_{moon} \left(\frac{2}{r_p} - \frac{1}{a} \right)}$ which equates to 2.448 km/s.

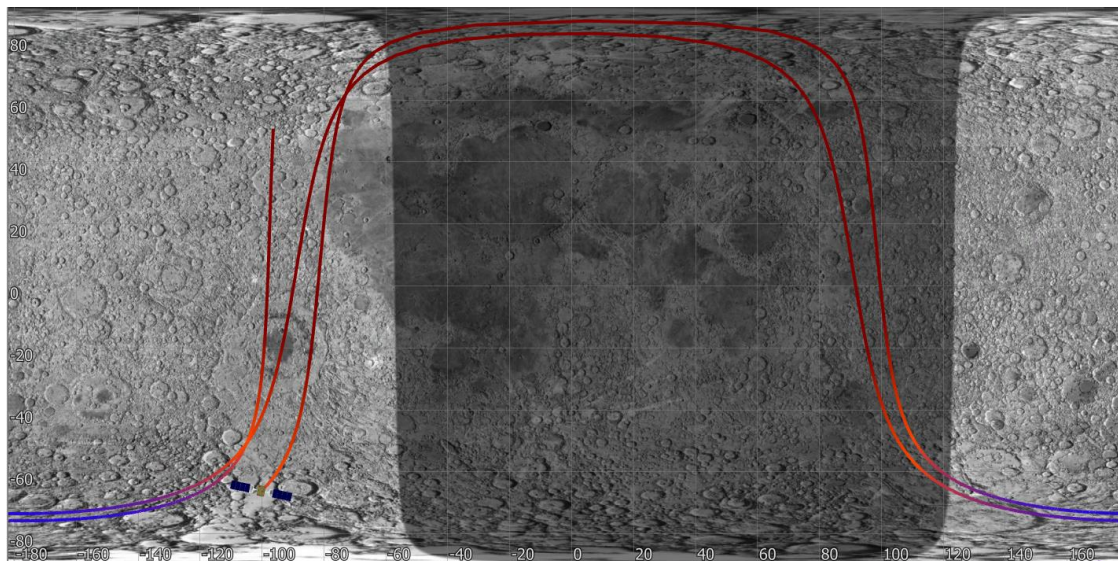


Figure 1: Ground Track over the Moon of the Communication Satellite

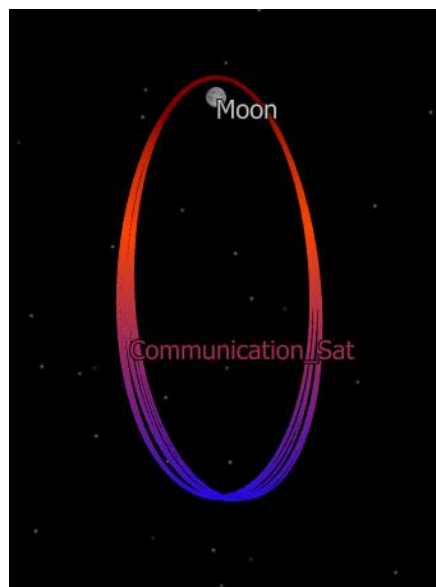


Figure 2: Orbit of Communications Satellite as Viewed from Earth

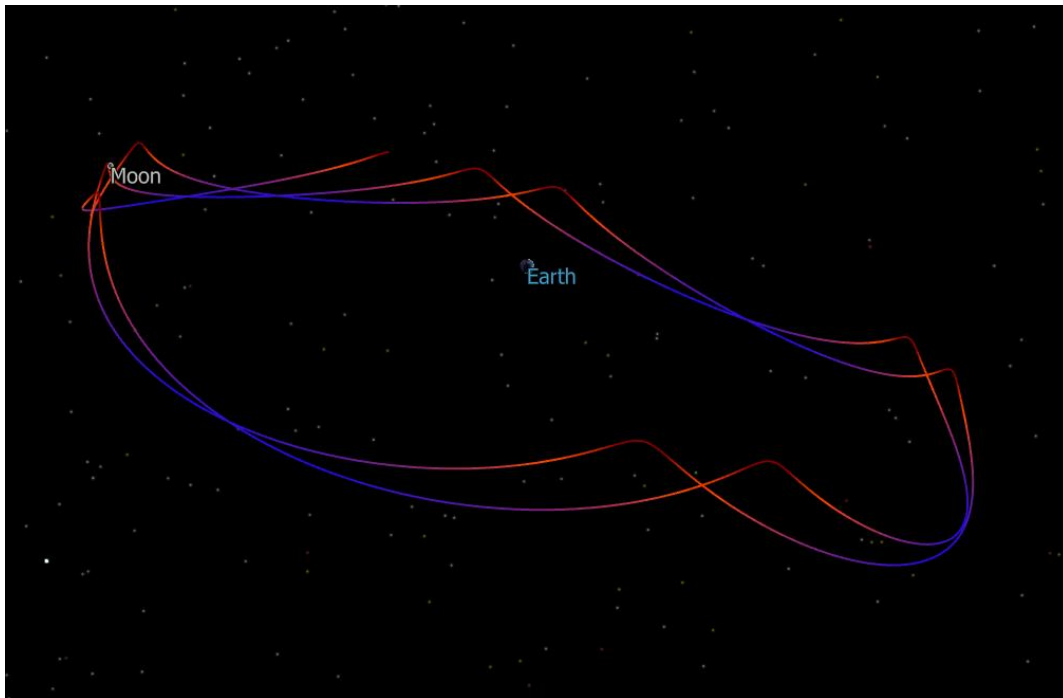


Figure 3: Earth-Centered Inertial View of the Orbit

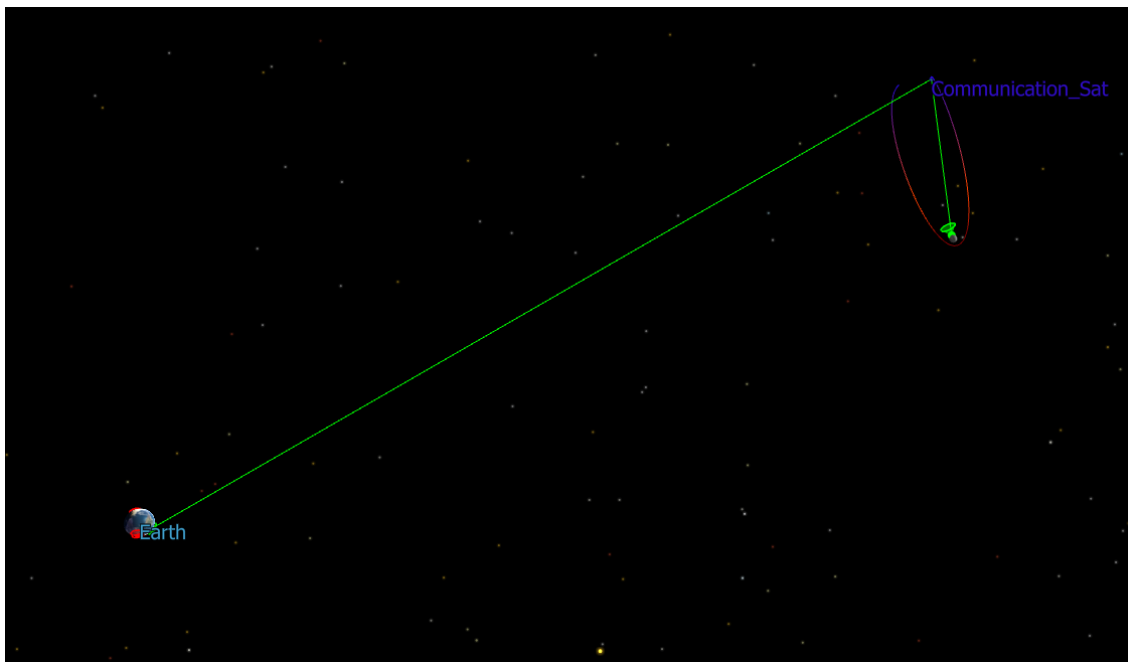


Figure 4: Example of Communication Geometry

Orbital perturbations do not affect the choice of orbit, as an NRHO is a very stable orbit between L1 and L2 that requires little correction. The communications satellite would require some

correction, as the orbit is not perfectly stable and would depart NRHO if unmaintained for several months (Davis, McCarthy, & M, 2022). It is also susceptible to any orbital perturbation that is caused by itself such as accidental thrust or exhaust venting.

There would be three ground stations on the Earth such there is always communication with at least one of the stations due to the Earth's rotation. These would be in: Cape Canaveral, FL; Falwell, United Kingdom; and Guam. These locations should guarantee communication capabilities for more than 90% of the day because they are spaced out roughly in longitudinal "thirds" and therefore two ground stations should always be visible to the orbiter.

The orbiter and the lander will both be exposed to ionizing radiation from both solar wind and solar flares. To account for this, they must either be properly shielded against the radiation, or more likely, have periodic restarts to account for potential single event effects. The rover would also be subject to the abrasive nature of the lunar dust. It would have to be designed to be able to withstand the dust in moving parts and on sensors for the duration of the mission.

Any launch vehicle capable of reaching the moon would be acceptable, though the SLS could be the best vehicle given that it is already planned to go to NRHO through the Artemis program and as such has a clear demonstrated ability for this exact purpose. For the same reason, Starship could also be a good option. The biggest consideration to a launch vehicle is the ability for the vehicle to set up the spacecraft for a lunar insertion at the lunar north pole so that the desired orbit can be achieved.

Section 4: Systems Thinking

One key design decision for this mission was the selection of the orbital architecture for the communications relay satellite that will transmit data between the lunar rover and Earth. Several orbit options were considered, including a Low Lunar Orbit (LLO), a polar elliptical orbit, and a Near Rectilinear Halo Orbit (NRHO). These options were evaluated based on communication coverage of the lunar south pole, orbital stability, propellant requirements, and compatibility with existing lunar exploration infrastructure. While LLO would provide strong communication with Earth. A polar elliptical orbit could provide wider surface coverage but would still have limited communication windows and moderate station keeping requirements. NRHO was ultimately selected because it offers long duration visibility of the lunar south pole, relatively low station keeping requirements, and compatibility with NASAs Artemis architecture, making it the most reliable option for maintaining communication with both the rover and Earth throughout the mission.